

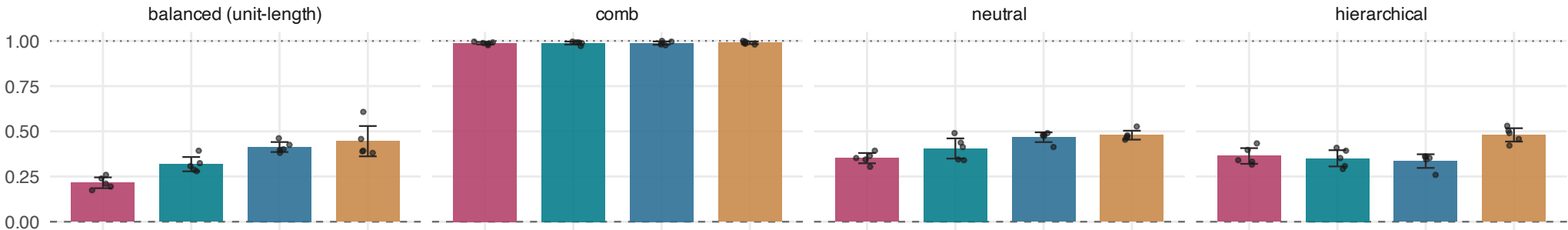
# Recorder comparison across tree shapes, capacity-controlled

Neighbour joining, 250 cells, 5 seeds. BASELINE 2x50, WT-CRISPR 13x10, PEtracer 14x3 (9-state), PALINCODE 30x2

A

## Topology error (normalised RF)

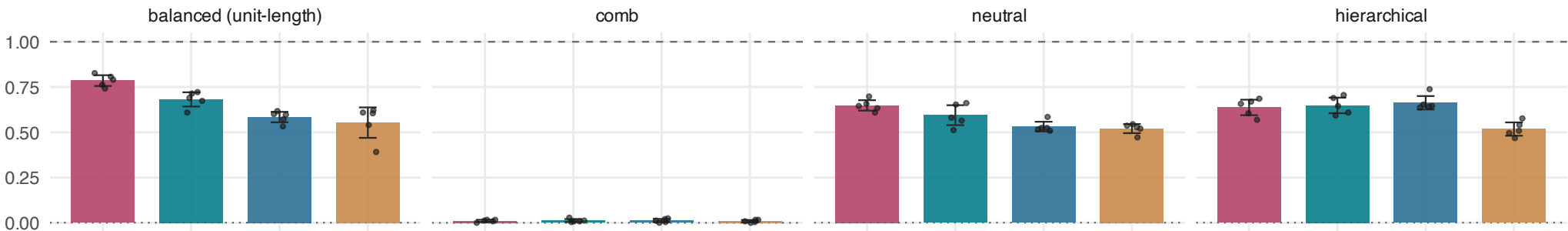
lower is better; dashed = perfect recorder, dotted = random



B

## Clade recall

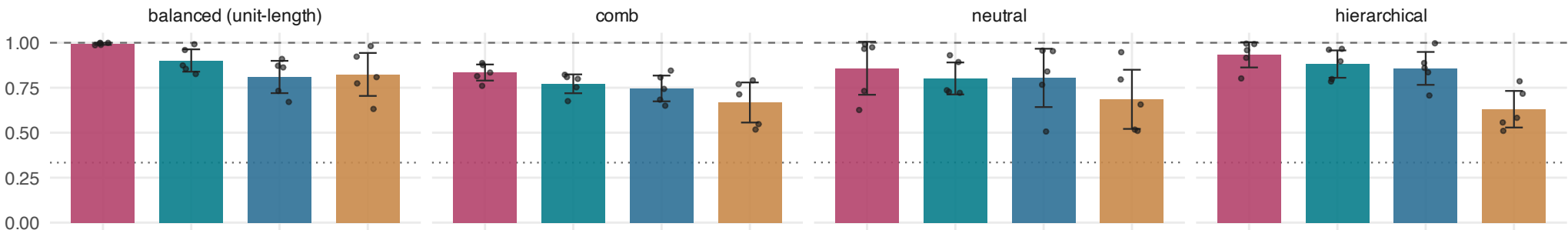
higher is better



C

## Rooted triplet agreement

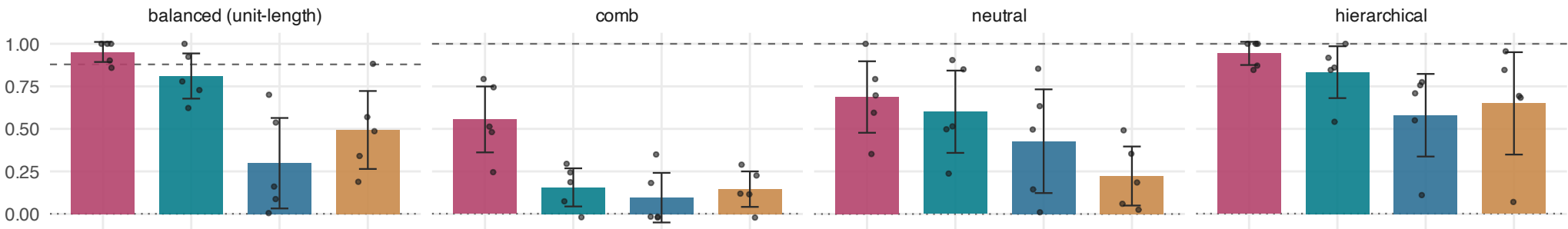
higher is better; chance is 0.33



D

## Clone assignment, k = 5 (ARI)

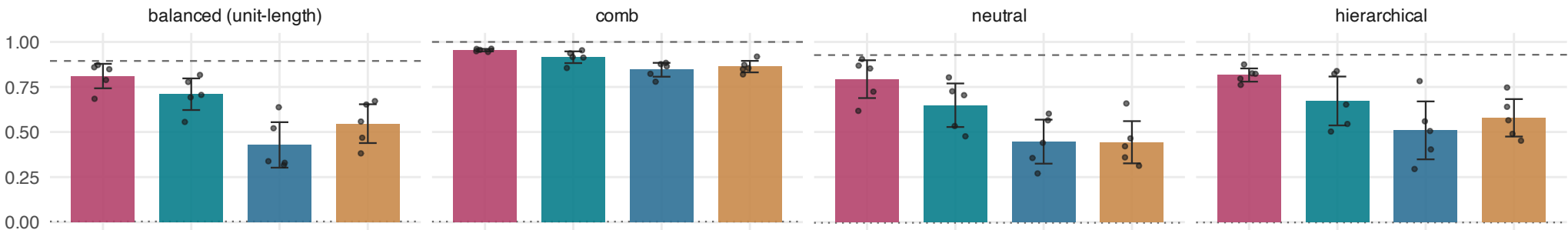
higher is better



E

## Cophenetic rank correlation

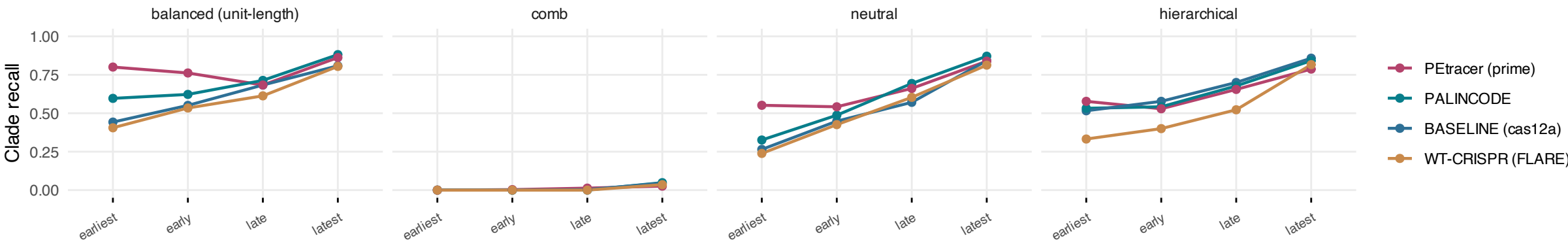
higher is better; the perfect reference sits at 0.93, not 1.00



F

## Which splits are recovered: ancient or recent

True clades binned by when they arose; a flat line means deep structure survives



Capacity is controlled, not matched: BASELINE carries 17-58% more bits than PETracer in every shape, so PETracer's wins come from a capacity deficit. Comb is the case where a single metric misleads -- no recorder recovers its clades (RF ~0.99, recall ~0.01) yet cophenetic correlation reaches 0.85-0.95, so relative distances are largely right even though the topology is not. Error bars are 95% intervals over 5 seeds and points are the seeds themselves: on hierarchical BASELINE, PALINCODE and PETracer are statistically indistinguishable (paired  $p = 0.45$  and  $0.53$ ), and on comb all four are tied at failure, so only balanced and neutral separate the recorders -- there PETracer beats every other recorder on paired tests ( $p = 0.001$ - $0.046$ ).